

CAH Triplets & Cousins — 21, 11, 17 Hydroxylase and Beyond



1. Mnemonic: 21 salt / 11 pressure / 17 no sex

WHAT YOU SEE

Mnemonic board: 21 loses salt, 11 has pressure, 17 has no sex

WHAT IT ENCODES

Master rule for CAH: 21-hydroxylase deficiency = salt wasting; 11-beta-hydroxylase = hypertension; 17-alpha-hydroxylase = no sex steroids. All share cortisol block with high ACTH and adrenal hyperplasia.

BOARD TRAP

Mixing up which enzyme causes HTN vs salt-wasting — 21 wastes salt, 11 raises pressure.

2. 17-OHP is the master marker

WHAT YOU SEE

Bottom banner: 17-OHP is the master marker

WHAT IT ENCODES

Elevated 17-hydroxyprogesterone is the screening/confirmatory marker for classic 21-hydroxylase deficiency, the most common CAH (~90%). It accumulates just above the enzyme block.

BOARD TRAP

17-OHP is diagnostic for 21-OH deficiency, NOT for 11 or 17 hydroxylase — don't apply it to the cousins.

3. 21-OH baby: salt wasting

WHAT YOU SEE

Infant labeled 21-OH baby, salt wasting, ambiguous genitalia

WHAT IT ENCODES

Classic 21-hydroxylase deficiency presents in neonates with salt-wasting crisis (hyponatremia, hyperkalemia, hypotension) and virilized/ambiguous female genitalia. Low aldosterone + low cortisol, high ACTH, high androgens.

BOARD TRAP

Salt-wasting newborn with hyperkalemia + hyponatremia = 21-OH; do not confuse with pyloric stenosis (which gives hypochloremic alkalosis).

4. 21-OH: low aldosterone, high K

WHAT YOU SEE

Dials: aldosterone low, renin high, K high

WHAT IT ENCODES

In 21-OH deficiency, aldosterone is low so renin rises and potassium climbs (hyperkalemia). Volume depletion and hypotension follow. Treat with glucocorticoid + mineralocorticoid (fludrocortisone) + salt.

BOARD TRAP

High renin + hyperkalemia distinguishes salt-wasting 21-OH from the hypertensive 11/17 forms (which have low renin).

5. Nonclassic 21-OH toddler/teen

WHAT YOU SEE

Center figure: nonclassic, mild, presents later with androgen excess

WHAT IT ENCODES

Nonclassic 21-OH deficiency presents later (childhood–young adult) with hirsutism, acne, precocious puberty, or PCOS-like infertility — not salt wasting. Cortisol is adequate; only androgens are mildly high.

BOARD TRAP

Nonclassic 21-OH mimics PCOS — screen with 17-OHP; it does NOT cause salt-wasting crises.

6. 11-beta-OH: hypertension

WHAT YOU SEE

Teen with BP high, K low, 11 has pressure

WHAT IT ENCODES

11-beta-hydroxylase deficiency blocks cortisol/aldosterone but accumulates 11-deoxycorticosterone (DOC), a mineralocorticoid → hypertension and hypokalemia. Androgens still high → virilization.

BOARD TRAP

11-OH causes HYPERTension with low K (from DOC), opposite of 21-OH's salt wasting — and still virilizes.

7. 11-OH: DOC drives HTN

WHAT YOU SEE

Label: 11-deoxycorticosterone has mineralocorticoid build-up

WHAT IT ENCODES

The hypertension in 11-beta-OH deficiency comes from DOC (11-deoxycorticosterone), which has aldosterone-like activity. This suppresses renin and aldosterone yet still retains salt.

BOARD TRAP

HTN here is DOC-mediated, not aldosterone — aldosterone is actually low, distinguishing it from primary aldosteronism.

8. 17-alpha-OH teen: no sex steroids

WHAT YOU SEE

Figure: 17-OH teen, androgens/estrogens low, BP high, K low

WHAT IT ENCODES

17-alpha-hydroxylase deficiency blocks both cortisol and all sex steroids → no puberty/ambiguous-to-absent secondary sex characteristics, plus HTN and hypokalemia from DOC/corticosterone excess.

BOARD TRAP

17-OH = hypertension + sexual infantilism (no sex steroids); 46,XY appear phenotypically female — opposite androgen picture from 21/11.

9. 17-OH: HTN + low K

WHAT YOU SEE

BP high, K low markers on 17-OH figure

WHAT IT ENCODES

Like 11-OH, 17-OH deficiency causes mineralocorticoid-excess hypertension with hypokalemia and suppressed renin, but uniquely also has absent androgens AND estrogens.

BOARD TRAP

Both 11 and 17 cause HTN/low K; 17 is set apart by ABSENT sex steroids (delayed puberty, primary amenorrhea).

10. 3-beta-HSD cousin

WHAT YOU SEE

Left figures: 3βHSD cousin, both sexes affected differently

WHAT IT ENCODES

3-beta-HSD deficiency blocks early steroidogenesis — low cortisol, aldosterone, and potent androgens. Males are undervirilized; females are mildly virilized by weak DHEA. Salt wasting can occur.

BOARD TRAP

3βHSD gives UNDERvirilized males (unlike 21/11 which virilize females) due to weak DHEA accumulation.

11. POR (cytochrome P450 oxidoreductase) cousin

WHAT YOU SEE

Right infant: POR cousin, skeletal malformations

WHAT IT ENCODES

P450 oxidoreductase deficiency impairs multiple CYP enzymes (21, 17, aromatase) → mixed CAH picture with skeletal malformations (Antley-Bixler), maternal virilization in pregnancy, and ambiguous genitalia in both sexes.

BOARD TRAP

POR deficiency is the CAH cousin associated with SKELETAL/craniofacial malformations — a unique distinguishing feature.

12. Cortisol block → high ACTH

WHAT YOU SEE

Subtitle: all have cortisol block, differ in what else

WHAT IT ENCODES

Every CAH variant shares a cortisol synthesis block, so ACTH rises and drives adrenal hyperplasia. What differs is the downstream pattern of mineralocorticoids and sex steroids above/below the block.

BOARD TRAP

All CAH = low cortisol + high ACTH + hyperplasia; the variants are told apart by salt, BP, and sex-steroid effects.

13. Forensic chemist (enzyme assay)

WHAT YOU SEE

Forensic chemist with test tubes panel

WHAT IT ENCODES

Diagnosis is by measuring the specific steroid that piles up above each block: 17-OHP for 21-OH, 11-deoxycortisol/DOC for 11-OH, and low sex steroids with high DOC for 17-OH.

BOARD TRAP

Each enzyme block has its own signature precursor — match the elevated steroid to the deficient enzyme.

14. Mineralocorticoid/androgen key table

WHAT YOU SEE

Top-left key table of BP, K, mineralocorticoid, androgen markers

WHAT IT ENCODES

Quick grid: 21-OH (low MC, low aldo, high K, high androgen); 11-OH (high MC via DOC, HTN, high androgen); 17-OH (high MC, HTN, low androgen). Use BP + K + androgen to localize.

BOARD TRAP

Read three axes together — BP, potassium, and androgen — to nail the enzyme; any single value can mislead.